Multimodal Skin Lesion Classification and Screening with an Improved EfficientNetV2 Ensemble



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UNDERTAKING

We, **Aachintya Wairiya (20233001)**, **Aadi Jain (20233002)**, **Aryan Jadhao (20233084)** and **Ansh Shekokar (20233061)**, submit this report entitled **"Multimodal Skin Lesion Classification and Screening with an Improved EfficientNetV2 Ensemble"** to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering.

This report documents our project implementation, experimental analysis and demonstration. Prior research, datasets and reused software components are to be acknowledged through the references and implementation credits. We accept responsibility for checking the accuracy of the content and for identifying our contributions separately from the work of others before submission.

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CERTIFICATE

This is to certify that the project report entitled **"Multimodal Skin Lesion Classification and Screening with an Improved EfficientNetV2 Ensemble"**, submitted by **Aachintya Wairiya (20233001)**, **Aadi Jain (20233002)**, **Aryan Jadhao (20233084)** and **Ansh Shekokar (20233061)** to Motilal Nehru National Institute of Technology Allahabad, Prayagraj, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of the project work undertaken under my guidance and supervision.

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PREFACE

This project examines how clinical photographs, dermoscopic images and patient metadata can be combined for skin lesion classification and broader screening. The work brings together image preprocessing, transfer learning, multimodal fusion and interpretable model outputs within a single experimental pipeline.

The report describes the MILK10k dataset, the specialist EfficientNetV2 image models, the metadata classifier and the feature and prediction fusion approach. It also discusses evaluation measures, category-level performance and the Streamlit demonstration developed to present model outputs and specialist heatmaps in an accessible form.

Through this project, we have gained practical experience in preparing paired image data, working with trained neural networks, comparing fusion strategies and communicating experimental findings. The report records the design decisions, implementation and limitations of the system, together with directions for further development.

ACKNOWLEDGEMENT

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ABSTRACT

Skin lesion classification draws on several sources of evidence: clinical appearance, dermoscopic structure and patient context provide complementary evidence, while severe class imbalance can conceal diagnostic errors behind overall accuracy. We present LesionLab, a multimodal EfficientNetV2-S system that combines these three sources and tests whether fusion can be improved without retraining the image specialists.

Our initial system combines the class probabilities of clinical-image, dermoscopic-image and metadata specialists through logistic-regression fusion. We enhance this system with two complementary prediction paths. The first uses a class-balanced classifier over a 173-dimensional representation containing PCA-compressed image embeddings, encoded metadata and specialist probabilities. The second averages image predictions across six geometric views. Equal-weight geometric pooling combines the two paths, using visual representations and variation across image views without changing the trained specialist weights.

Experiments use the MILK10k release containing 5,240 lesions, 10,480 images arranged in clinical–dermoscopic pairs, and eleven diagnostic categories. On the same 786-lesion internal validation cohort, the enhanced ensemble increases macro ROC-AUC from 0.8627 to 0.8767, macro average precision from 0.4560 to 0.5576, and mean class Dice from 0.3551 to 0.553. Mean one-versus-rest class accuracy increases from 0.9267 to 0.9440, macro sensitivity from 0.4282 to 0.5430, and macro specificity from 0.9569 to 0.9638. These metrics follow the terminology of the JI-ADF reference while retaining an explicit argmax decision rule for our models.

A Streamlit interface connects the ensemble to input-relevance checks, modality comparisons, uncertainty summaries and specialist-level Grad-CAM visualizations. On this validation cohort, feature integration and view averaging improve on our initial score-fusion system. Independent evaluation would further assess generalization.

Keywords: multimodal learning; MILK10k; EfficientNetV2; frozen features; test-time augmentation; broader lesion screening.

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Abbreviations and Key Terms

Term	Meaning
JI-ADF	Joint-Individual Learning with Adaptive Decision Fusion; the methodological reference.
MLP	Multilayer perceptron, used for patient metadata.
PCA	Principal-component analysis, fitted on training image features.
TTA	Test-time augmentation: averaging predictions across image transformations.
OOF	Out-of-fold predictions from rows excluded from a fold's downstream fit.
CLAHE / ROI	Contrast-limited adaptive histogram equalization / region of interest.
Dice	Class-wise overlap of predicted and reference positives; averaged equally over eleven categories.
Sensitivity	Per-class true-positive rate, averaged equally over diagnoses.
Specificity	Per-class true-negative rate, averaged equally over diagnoses.
ROC-AUC	Area under the one-versus-rest receiver-operating-characteristic curve.
AP / PPV / NPV	Average precision / positive predictive value / negative predictive value.
Broader screening	Malignant/precancerous-family versus benign/inflammatory-family grouping, rather than confirmed malignancy.
Grad-CAM	Gradient-based spatial explanation of an individual image classifier.

CHAPTER 1

INTRODUCTION

A skin-lesion classifier must identify relevant features in photographs, measurements or patient records to predict a diagnosis. In this project, each lesion has two photographs: a clinical close-up and a dermoscopic image. The accompanying metadata describe age, sex, anatomical site and a release-specific skin-tone category. MILK10k supplies these paired records and eleven diagnostic labels [1].

We developed LesionLab around this paired-input setting in two stages. Stage 1 was our initial multimodal system: we implemented and adapted two EfficientNetV2-S image classifiers, a metadata multilayer perceptron and logistic-regression fusion of their class scores. Stage 2 enhanced that system: we retained our trained specialists and investigated a richer decision layer, combining a classifier over their frozen visual features and metadata with predictions averaged over six image views. The same work also produced input rejection, a Streamlit interface, branch comparisons, inconclusive-result messages and image heatmaps.

1.1 Problem and Motivation

A photograph may show a lesion poorly, the two views may favour different diagnoses, and the available metadata may be incomplete. A well-formed photograph may also show an unrelated object. A classifier trained only to choose between eleven lesion categories will still choose one when shown such an object. Input relevance therefore requires a separate check before diagnosis inference.

Class imbalance creates another problem. In the selected release, BCC accounts for 2,522 of 5,240 lesions, while MAL_OTH has nine. A model can improve total accuracy by handling the dominant category better and still fail on a small category. We therefore evaluate class-averaged discrimination, sensitivity, specificity, predictive values and Dice alongside mean one-versus-rest class accuracy. These measures answer different questions; none can replace all the others.

The first demonstration also exposed a communication problem. A list of eleven percentages does not explain whether the specialists agree or whether two categories have nearly equal scores. It can make a weak preference appear more decisive than the evidence supports. We added summaries that describe the recorded scores and branch agreement. They do not invent visual findings or ask a language model to generate a diagnosis.

The model experiments test whether the decision layer can be improved without replacing the trained image encoders. Each specialist already produces a 1,280-dimensional image embedding, but the initial score-fusion system uses only its eleven output scores. The feature-based extension gives the downstream classifier access to a compressed version of those embeddings. A second member averages predictions across image transformations. Their combination tests whether these two summaries contain useful complementary information.

1.2 Objectives and Contribution

Our objective is to develop a multimodal classifier and evaluate the enhancement. Stage 1 establishes our initial score-fusion system; Stage 2 adds the feature/TTA ensemble. We compare these two stages of our project on identical lesion records. We keep the class order, reference labels and trained specialists fixed. The comparison uses the 786-lesion validation partition; the historical 524-lesion holdout experiment remains a separate result.

After implementing, adapting and training the multimodal score-fusion system, we developed and evaluated four outputs in the enhancement stage. First, it provides a 173-dimensional representation containing two compressed image embeddings, encoded metadata and three specialist score vectors. Second, it evaluates a fixed geometric combination of the feature classifier and six-view image averaging. Third, it documents which diagnosis errors improve and which remain. Fourth, it connects the selected method to a working interface with input checks, comparisons and branch-level explanations.

Our contribution is the implementation and experimental study of this combination. EfficientNetV2, PCA, logistic regression, test-time averaging and Grad-CAM are established methods. We do not claim to have invented those components or to have produced a new image backbone. We assembled, checked and evaluated them for the paired MILK10k task.

1.3 Research Questions

The report addresses the following questions.

1. Does the frozen-feature/TTA ensemble improve the selected JI-ADF metric family relative to our initial score-fusion system on the same 786 lesions?
2. Which diagnostic categories and confusion patterns account for the change in aggregate performance?
3. How do our internal-validation measurements relate to the published JI-ADF values under different evaluation protocols?
4. Does the interactive inference path reproduce the preserved implementation, and can the interface explain its outputs without changing the predictions?

These questions separate model quality from software correctness. A correct inference implementation can reproduce an incorrect diagnosis. Conversely, an attractive interface or a plausible heatmap does not establish that a prediction is correct.

1.4 Scope

The main study uses one MILK10k release: 10,480 image rows from 5,240 lesions. Each image pair stays in one partition. Patient identifiers are unavailable, so the release cannot establish patient-level independence. The main comparison uses a separate 786-lesion validation partition. Appendix A records the design of the separate direct-binary study. Its performance results are outside the selected multiclass evaluation scope.

CHAPTER 2

RELATED WORKS

2.1 EfficientNetV2 and Image Transfer Learning

Tan and Le introduced EfficientNetV2 as a convolutional family designed around training speed and parameter efficiency [2]. Their architecture search includes Fused-MBConv operations, and the paper also studies progressive learning. LesionLab uses the EfficientNetV2-S architecture through torchvision with ImageNet initialization. It does not reproduce the architecture search or the paper's progressive image-size schedule.

In our implementation, the backbone's original classifier is removed. Each image branch then applies dropout and a linear eleven-class head to the pooled representation. Clinical and dermoscopic images use separate instances and separate trained weights. This permits different preprocessing and augmentation policies while preserving the same embedding dimension. Sharing an architecture name therefore does not mean the two branches learn the same decision function.

The original specialist recipe includes Mixup, label smoothing and AdamW [3–5]. These choices regularize training, but the project did not run a controlled removal experiment for every choice. Their presence is a configuration fact, not evidence that each independently improved the final result. The feature/TTA extension uses the resulting frozen specialists and leaves their regularization history unchanged.

2.2 JI-ADF as the Multimodal Reference

JI-ADF combines clinical photographs, dermoscopic images and metadata through joint and individual prediction streams [6]. Auxiliary heads supervise the individual evidence, while a learned gate assigns sample-dependent weights to the branch outputs. Its attention module exchanges image and metadata information and includes a residual path. The reference therefore studies both representation interaction and decision weighting.

Our local adaptation implemented this architectural direction as a separate experiment. Its measured scores describe that implementation and training configuration. They are not the paper's published benchmark scores. Differences in input preparation, fitting data, optimization and model selection prevent a shared backbone from making two experiments equivalent. This distinction limits comparisons between a complex joint model and a modular late-fusion system.

The reference led us to test whether a downstream learner could use the available representations without fitting a joint image encoder again. This alternative reduces the amount of neural training needed to test a new fusion method. It also lets the original specialists remain independently inspectable.

2.3 Score Fusion and Feature Fusion

Score fusion starts from the existing specialists' predictions. With three eleven-class specialists, the original stack has 33 inputs. A downstream classifier can learn that one branch is more useful for a category or that certain score combinations recur. It cannot reconstruct image information that all the output heads have discarded.

Feature fusion supplies image embeddings to the downstream classifier. The image embeddings retain many more coordinates than the eleven-score outputs. Directly concatenating both 1,280-dimensional vectors would create a relatively large downstream input for a small fitting cohort. We therefore compress each branch with PCA and retain its specialist scores as additional information. PCA itself uses no diagnosis labels; the subsequent classifier learns the supervised decision boundary [7].

Table 2.1. Fusion strategies and the information supplied to their decision layers.

Strategy	Input to the decision layer	What changes in this project
Initial score-fusion system	Three eleven-class score vectors	Class-balanced logistic regression on 33 scores.
Frozen-feature classifier	Two PCA embeddings, metadata and specialist scores	A newly fitted downstream boundary with fixed image encoders.
Six-view image averaging	Clinical and dermoscopic predictions across six views	Additional inference passes; no new image weights.
Feature/TTA ensemble	The feature classifier and image-average distributions	A fixed geometric combination, not a learned sample gate.

The methods differ in fitting access as well as representation. The initial score-fusion system fits on the fusion partition. The feature learner can also use training-lesion representations. The matched evaluation cohort makes their predictions comparable, but does not isolate the causal effect of the geometric pooling equation. A matched-label-budget study would be needed for that narrower claim.

2.4 Test-Time Prediction Averaging

Test-time augmentation combines predictions obtained from transformed versions of an input [8]. LesionLab uses identity, horizontal flip, vertical flip and three right-angle rotations. It averages views within an image branch and then averages the two image modalities. It does not learn an augmentation controller or choose a transformation after seeing the reference diagnosis.

Averaging can reduce dependence on orientation, but it requires extra forward passes and may preserve a systematic mistake across all views. The experiment therefore compares the image average with the feature classifier and their combination, rather than treating more inference passes as automatic evidence of improvement.

CHAPTER 3

PROPOSED WORK

3.1 System Design

LesionLab separates the image models from the decision layer and the interface. The image specialists produce embeddings and class scores. A saved metadata encoder converts a patient record into the numeric input expected by the metadata model. The ensemble combines these outputs, and the interface explains the resulting prediction. A relevance gate runs before this sequence for interactive uploads.

This separation helped us investigate errors without retraining immediately. We could replay an uploaded case through the initial score-fusion system, compare the preprocessing output, inspect each branch and then test another downstream classifier. The same design allows an unsuccessful fusion experiment to remain separate from the working demonstration. It also identifies the artifacts used for each prediction.

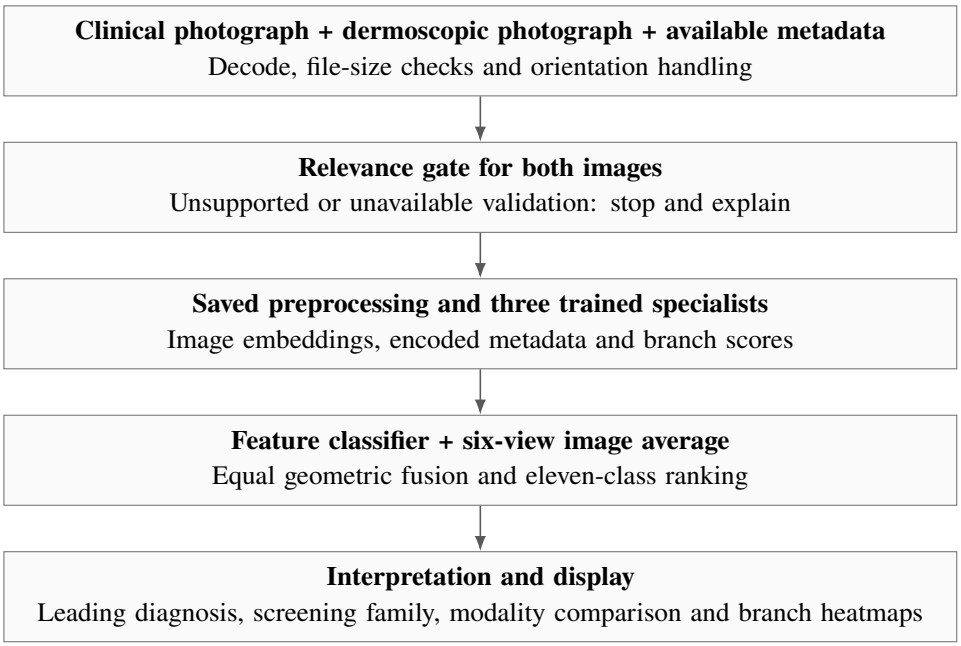


Figure 3.1. Interactive inference sequence. Failure at the input gate stops disease processing; explanation is attached after a valid prediction.

The main path requires a clinical and a dermoscopic image of the same lesion. Metadata can be partly missing because the saved encoder handles missing and unknown values. The paired ensemble does not silently duplicate one photograph to manufacture a missing modality. Single-image modes use one specialist and are labelled separately in the interface.

3.2 Dataset and Exploratory Analysis

The selected MILK10k release contains 10,480 image rows from 5,240 lesions. Each lesion contributes a clinical close-up and a dermoscopic image. A lesion identifier links the two rows, the shared metadata

and the diagnosis. The ingestion checks confirm two supported image types per lesion, consistent labels within each pair and no lesion identifier assigned to more than one partition.

Table 3.1. Canonical diagnosis order and lesion counts in the selected release.

Code	Diagnostic family	Lesions
AKIEC	Actinic keratosis / intraepidermal carcinoma	303
BCC	Basal cell carcinoma	2522
BEN_OTH	Other benign proliferations	44
BKL	Benign keratinocytic lesion	544
DF	Dermatofibroma	52
INF	Inflammatory and infectious conditions	50
MAL_OTH	Other malignant proliferations	9
MEL	Melanoma	450
NV	Melanocytic nevus	746
SCCKA	Squamous cell carcinoma / keratoacanthoma	473
VASC	Vascular lesions and hemorrhage	47

BCC contributes 48.13% of all lesions. The largest category has about 280 times as many examples as MAL_OTH, and five categories have fewer than 100 lesions. Stratification preserves this imbalance rather than eliminating it. For example, validation contains 378 BCC lesions and only one MAL_OTH lesion. This is why the report always gives a support count beside a category-specific score.

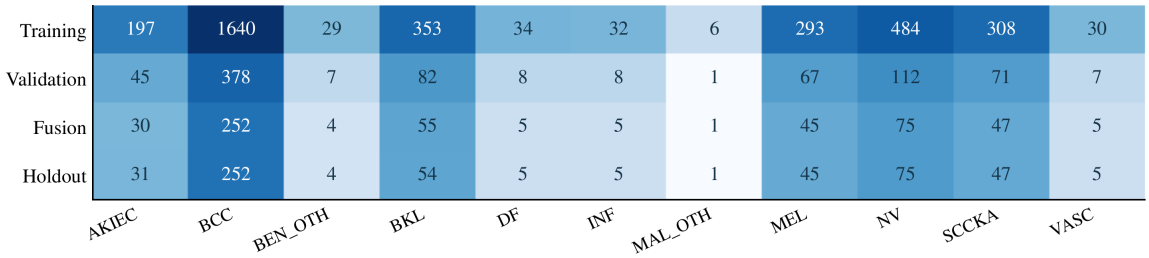


Figure 3.2. Lesion support by diagnosis and partition. Annotations are counts; colour uses a logarithmic scale.

The data are partitioned by lesion into 3,406 training, 786 validation, 524 fusion and 524 holdout cases. Keeping both photographs together prevents an image pair from being split between fitting and evaluation. The available release does not provide usable patient identifiers for these partitions. Two lesions from the same person could therefore remain in different partitions; the present checks cannot resolve that possibility.

Metadata were summarized once per lesion, not once per photograph. Recorded age is missing for 20 lesions and anatomical site for 31, corresponding to 0.38% and 0.59%. Median age is 65 years, with an interquartile range of 50–75. The release includes 3,152 male and 2,088 female records. Skin-tone category 3 contains 3,174 lesions, or 60.57% of the cohort; category 0 contains six. These 0–5 categories are release-specific and are not treated as Fitzpatrick types.

All 10,480 distributed image headers were readable and reported 600-by-450-pixel dimensions. This check detects absent or malformed files, not blur, framing quality or whether the two photographs really show the intended lesion. Those are different aspects of data quality. The application uses additional file checks for uploaded images, while a trained gate addresses relevance.

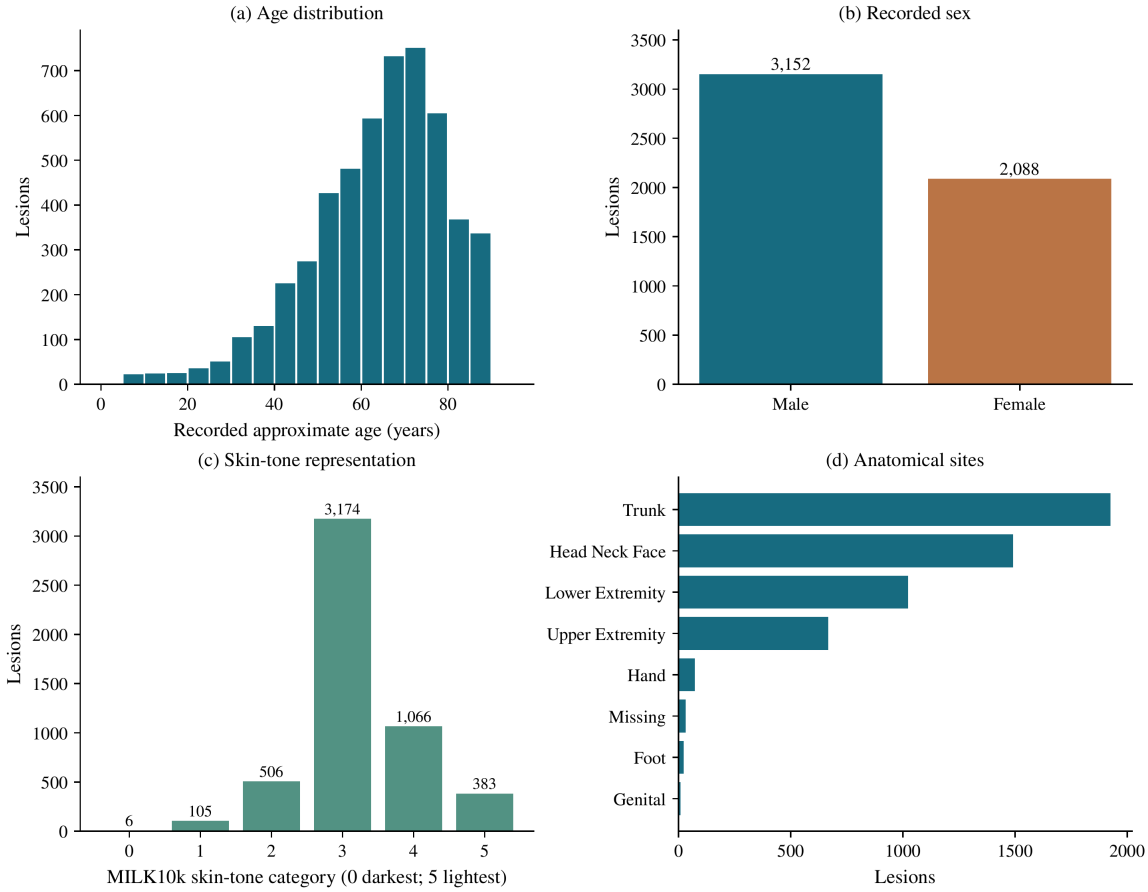


Figure 3.3. Lesion-level metadata distributions. These describe the available records; they do not establish equal classifier performance across demographic groups.

3.3 Preprocessing

The training and demonstration paths use the same modality-specific preprocessing functions. Images are decoded as RGB. Dermoscopic images first pass through a hair-mask and inpainting stage; clinical images skip it. Both branches then apply luminance CLAHE, a heuristic foreground crop, smoothing, resizing and the saved normalization transform.

Table 3.2. Preprocessing operations in the preserved specialist implementation.

Operation	**Recorded setting**
Hair mask, dermoscopy only	Grayscale black-hat operation; 21-by-21 rectangular kernel; threshold 20.
Hair inpainting	Telea inpainting with radius 3.
Contrast enhancement	CLAHE on LAB luminance; clip limit 2.0; 8-by-8 tile grid.
Foreground crop	LAB colour-channel thresholding and morphology; accepted area ratio 0.02–0.95; padding ratio 0.05.
Final geometry	3-by-3 median filter and bicubic resizing to 224 by 224 pixels.
Tensor normalization	ImageNet mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$.

The crop is not a learned segmentation model. It proposes a foreground region using colour and morphology and retains the original image if the region fails plausibility checks. No pixel-level ground-

truth mask is used. Consequently, the system cannot claim segmentation accuracy or a segmentation Dice score from this operation.

Training adds geometric transformations after deterministic preprocessing. The standard clinical policy uses horizontal and vertical flips and rotations up to 30 degrees. The dermoscopic policy uses horizontal flips and rotations up to 15 degrees. Stronger rare-class augmentation exists in the code, but the completed specialist configurations set it to false. The reported configuration records the options actually used to produce the saved models; unused options are not treated as improvements.

3.4 Metadata Representation

The encoder uses two numeric fields, age and skin-tone category, plus categorical sex and site. It fits numeric means, standard deviations and category vocabularies on training records. The saved vector contains two scaled numeric values, two sex indicators and eight site indicators, for twelve features in total. The site vocabulary includes an explicit unknown category.

The demonstration reuses this encoder. An omitted field remains missing until the fitted transform handles it; the interface does not fill it with the values from a known demonstration case. Diagnosis text, pathology confirmation, lesion identifiers and acquisition attribution are not predictors. In particular, identifiers are used for joining and partition checks, not as shortcut inputs to the classifier.

Table 3.3. Inputs to the three specialists and the downstream feature classifier.

Component	Representation
Clinical specialist	RGB tensor of shape $3 \times 224 \times 224$; 1,280 pooled features and eleven logits.
Dermoscopic specialist	The same tensor shape, with dermoscopy-specific preparation and separate weights.
Metadata specialist	Twelve encoded fields; hidden layers of 128 and 64 units; eleven logits.
Downstream classifier	64 clinical PCA coordinates, 64 dermoscopic coordinates, twelve metadata features and 33 specialist scores.

3.5 Specialist Models and Training

Each image branch starts from an ImageNet-initialized EfficientNetV2-S. We replace the original classifier with dropout at 0.4 and a linear eleven-class layer. The metadata network follows the sequence $12 \rightarrow 128 \rightarrow 64 \rightarrow 11$, with batch normalization, ReLU and dropout at 0.3 in its hidden stages. Every branch uses the canonical category order in Table 3.1.

The image schedules begin with a frozen-backbone phase and then unfreeze the backbone at a lower or equal learning rate. Sampling and loss weighting both use inverse-square-root class frequencies. Mixup strengths are 0.4 for clinical and 0.25 for dermoscopic images; label smoothing is also part of the specialist recipe. These operations affect training batches, not validation images.

The completed epoch counts are below the configured maxima because the runs use early stopping. A completion flag means that the scheduled run finished; it does not mean the final epoch supplied the selected checkpoint. The history files retain each epoch's validation loss and F1, so checkpoint selection can be distinguished from the last logged state.

Table 3.4. Configuration of the completed specialist runs.

Setting	Clinical	Dermoscopy	Metadata
Maximum epochs	30	20	20
Completed epochs	27	18	9
Batch size	32	32	64
Frozen-backbone epochs	6	4	—
Head learning rate	10^{-4}	10^{-4}	10^{-3}
Backbone learning rate	5×10^{-5}	10^{-4}	—
Early-stopping patience	5	5	5
Selection criterion	Macro F1	Macro F1	Macro F1

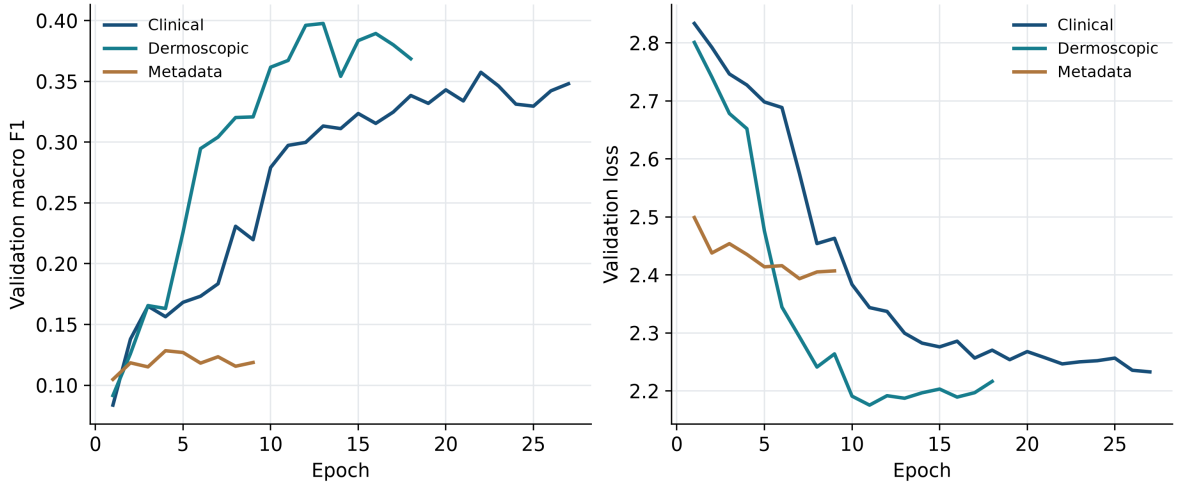


Figure 3.4. Preserved specialist histories. The curves are taken from the completed runs; no new image training was performed to create this report.

Kaggle provided two Tesla T4 GPUs. The workflow can run a separate image specialist on each device; it does not combine their memory into one larger GPU. Mixed precision and channels-last storage are enabled in the completed image configurations. Before a long run, a benchmark checks the actual runtime and candidate batch sizes. These engineering choices aim to reduce wasted GPU time; they do not establish separate accuracy improvements.

The workflow caches deterministic preprocessing so that later epochs do not repeat the OpenCV work for every image. Kaggle inputs remain read-only, while writable code, caches, logs and checkpoints live under the working directory. Artifact export records the trained state and processed split files. Restoring a run therefore requires matching both the model state and the split identity, rather than copying weights alone.

3.6 Initial Score-Fusion System and the Enhanced Ensemble

Stage 1 is our initial multimodal score-fusion system. We implement and adapt the specialist pipeline, train the clinical, dermoscopic and metadata branches, and combine their softmax vectors into 33 values. A class-balanced multinomial logistic regression with $C = 0.05$ fits the fusion partition. This initial system provides the internal reference for our subsequent development. Stage 2 is the enhanced feature/TTA ensemble: it changes the representation seen by a new downstream classifier

while leaving our trained specialists fixed. Both systems are stages of LesionLab; JI-ADF is a separate literature-inspired experimental comparison.

Let $\mathbf{f}_c$ and $\mathbf{f}_d$ be the pooled image embeddings, $\mathbf{m}$ the encoded metadata, and $\mathbf{p}_c, \mathbf{p}_d, \mathbf{p}_m$ the specialist score vectors. Separate PCA transforms compress each image embedding to 64 coordinates. The combined representation is

$$\mathbf{h} = [\mathrm{PCA}_{c,64}(\mathbf{f}_c); \mathrm{PCA}_{d,64}(\mathbf{f}_d); \mathbf{m}; \mathbf{p}_c; \mathbf{p}_d; \mathbf{p}_m] \in \mathbb{R}^{173}. \tag{3.1}$$

The dimensions sum to $64 + 64 + 12 + 33 = 173$. PCA fits only training-lesion embeddings. A standard scaler and class-balanced logistic regression with $C = 0.1$ then map this vector to an eleven-class distribution, denoted $\mathbf{q}^F$.

The downstream classifier is trained even though the image encoders are frozen. Its fitting labels include training and permitted fusion records. The extension therefore requires downstream training, but no additional image-backbone optimization during the ensemble study.

3.7 Six-View Averaging and Geometric Combination

For each prepared image, the second ensemble member applies identity, horizontal flip, vertical flip and rotations of 90, 180 and 270 degrees. The transformations operate on tensors without interpolating new image values. Predictions are averaged over the six clinical views and the six dermoscopic views, then combined with equal image-modality weights:

$$\mathbf{q}^T = \frac{1}{2}\left[\frac{1}{6}\sum_{t\in\mathcal{T}} \mathbf{p}_c(t(x_c)) + \frac{1}{6}\sum_{t\in\mathcal{T}} \mathbf{p}_d(t(x_d))\right]. \tag{3.2}$$

Metadata enter through the feature classifier; they are not a third member of this image average. Identity-view embeddings can be reused for the feature path. A paired prediction still requires twelve image forward passes, so this extension trades inference time for a richer decision.

The final score takes an equal geometric combination of $\mathbf{q}^F$ and $\mathbf{q}^T$ and normalizes over categories:

$$q_k = \frac{\sqrt{\max(q_k^F, \epsilon)\max(q_k^T, \epsilon)}}{\sum_{j=1}^{11}\sqrt{\max(q_j^F, \epsilon)\max(q_j^T, \epsilon)}}, \qquad \epsilon = 10^{-12}. \tag{3.3}$$

The floor prevents taking the logarithm of zero in the implementation. The geometric rule gives less support to a category when one member assigns it a very small score. It does not learn case-specific weights and does not guarantee that agreement is correct.

3.8 Input Rejection and Result Interpretation

The input path first checks whether a file decodes as an image and whether its dimensions meet the minimum size. The learned gate then estimates whether it is a supported lesion photograph. A missing validator, a nonfinite validator output or a score below its saved threshold stops disease processing. Both photographs must pass before paired inference starts.

The interpretation layer reads model outputs rather than generating new image findings. It flags a top-two score margin below 0.05, disagreement between image specialists, a combined category

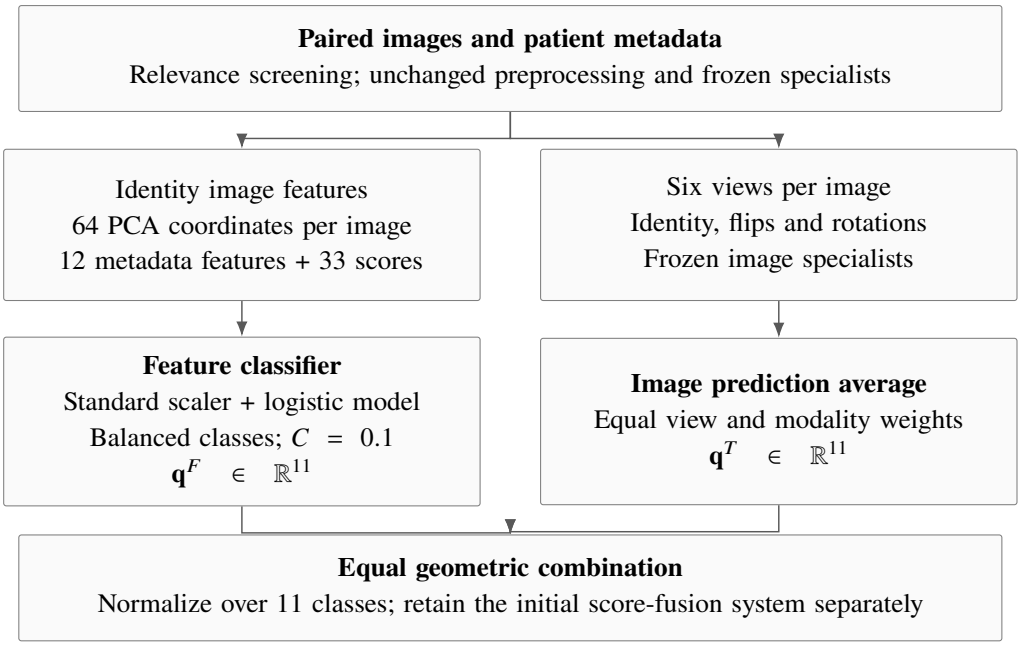


Figure 3.5. The two inference paths in the selected ensemble. Clinical, dermoscopic and metadata specialists retain their trained weights.

Table 3.5. Application states before and after disease inference.

State	Application response
Unreadable or undersized image	Explain the file problem; do not compute a disease prediction.
Unsupported image or unavailable gate	Withhold diagnosis scores and request a supported input.
Accepted pair	Run the saved models and retain all eleven category scores.
Disagreement or close leading scores	Display an inconclusive interpretation while preserving the original scores and leading category.
Unavailable heatmap	Keep the valid prediction and explain that the spatial explanation could not be produced.

preferred by neither image model, or disagreement between the two ensemble members. These are fixed display rules. They have not been fitted as a calibrated confidence model, and the report does not interpret the 0.05 margin as a clinically validated boundary.

3.9 Broader Screening Endpoint

For the interface, the positive family is $\mathcal{P} = \{\text{AKIEC}, \text{BCC}, \text{MAL_OTH}, \text{MEL}, \text{SCCKA}\}$; the other diagnoses form the benign/inflammatory family. The label follows the leading category:

$$\widehat{y} = \arg\max_k q_k, \qquad \widehat{b} = \mathbf{1}[\widehat{y} \in \mathcal{P}]. \tag{3.4}$$

The grouping includes precancerous and mixed categories rather than confirmed malignancy alone. It is an interface feature; its performance results are outside the multiclass evaluation in Chapter 4.

CHAPTER 4

EXPERIMENTAL SETUP AND RESULTS

4.1 Evaluation Protocol

We evaluate our initial score-fusion system and enhanced feature/TTA ensemble on the same 786 MILK10k internal validation lesions. Every lesion retains its reference diagnosis, class order and saved model scores. The aggregate metrics were calculated from the model predictions for this cohort using the definitions and decision rule below. The presentation of results uses eight measures also included in the JI-ADF paper: ROC-AUC, average precision, mean per-class accuracy, sensitivity, specificity, Dice, positive predictive value and negative predictive value [6].

The 786 validation lesions are excluded from PCA and downstream classifier fitting. The JI-ADF paper describes a separate official hidden-test cohort of 479 lesions. Its numbers provide published context, while the initial-versus-enhanced comparison uses identical local cases. Differences in cohort and hard-decision details prevent interpreting a numerical cross-study difference as controlled superiority.

4.1.1 Fitting Access and Selection

PCA fits 3,406 training-lesion embeddings without diagnosis labels. The enhanced feature classifier uses training labels and permitted fusion records, then refits on 3,930 training-plus-fusion lesions. The initial score-fusion system instead fits its score stack on 524 fusion lesions. Validation records enter neither PCA fitting nor downstream classifier fitting.

The study explored score-based, feature-based and neural fusion candidates before selecting the feature/TTA combination. Three-fold out-of-fold predictions on the fusion partition selected the final fixed combination. The matched results compare complete pipelines with different downstream fitting access rather than isolating the pooling equation alone.

4.2 Metric Definitions and Decision Rule

For each diagnosis k, the reference label is positive when the lesion belongs to that category. Our hard prediction assigns exactly one category, $\widehat{y}_i = \arg\max_j q_{ij}$. We then form a one-versus-rest confusion matrix for each of the eleven diagnoses. All categories receive equal weight in the macro average, including those with very low support.

4.2.1 Ranking Metrics

ROC-AUC measures ranking discrimination between a category and all other categories across score thresholds. We calculate each class AUC from its continuous score vector, then average the eleven values. Average precision summarizes the precision–recall ranking through recall-weighted precision increments. Macro average precision is the equal average of the class-specific values. It differs from PPV at a single hard decision rule.

The terminology follows Appendix C of JI-ADF. Its Table 1 labels 0.543 as precision, whereas Table 5 identifies 0.543 as average precision and separately gives mean PPV of 0.596. We use the more explicit Table 5 labels throughout this comparison [6].

4.2.2 One-versus-Rest Accuracy and Rates

For diagnosis k, let TP_k, TN_k, FP_k, FN_k denote the confusion counts and N the number of evaluated lesions. We compute

$$\mathrm{Acc}_k = \frac{TP_k + TN_k}{N}, \tag{4.1}$$

$$\mathrm{Sens}_k = \frac{TP_k}{TP_k + FN_k}, \qquad \mathrm{Spec}_k = \frac{TN_k}{TN_k + FP_k}. \tag{4.2}$$

Each aggregate is the arithmetic mean across the eleven classes. Mean per-class accuracy includes true negatives for every one-versus-rest task. It is not the fraction of lesions whose final eleven-class diagnosis is correct. A high value can coexist with weak recall for rare diagnoses, so it must be read alongside sensitivity and Dice.

4.2.3 Dice and Predictive Values

The class-level overlap and predictive-value metrics are

$$\mathrm{Dice}_k = \frac{2TP_k}{2TP_k + FP_k + FN_k}, \tag{4.3}$$

$$\mathrm{PPV}_k = \frac{TP_k}{TP_k + FP_k}, \qquad \mathrm{NPV}_k = \frac{TN_k}{TN_k + FN_k}. \tag{4.4}$$

We average these values equally across all diagnoses. Undefined ratios receive zero rather than removing a class from the average. For our argmax predictions, class Dice equals class F1 algebraically; we display the result under the paper's Dice terminology. This is classification overlap, not segmentation Dice. The application has no ground-truth pixel masks for evaluating crops or heatmaps.

The paper's full hard-decision and undefined-ratio conventions have not been established as identical to ours. Using the same metric names does not by itself establish protocol equivalence. Partial AUC at high sensitivity and expected calibration error also appear in the reference, but are not reproduced here because a matching calculation protocol has not been established.

4.3 Initial System and Enhanced Ensemble

Every selected aggregate improves over the initial score-fusion system on this validation cohort. Macro ROC-AUC increases from 0.8627 to 0.8767, while average precision increases from 0.4560 to 0.5576. These changes describe ranking of the eleven diagnoses using continuous model scores. They do not depend on an optimized per-class operating threshold.

Mean class Dice rises from 0.3551 to 0.553, and macro sensitivity from 0.4282 to 0.5430. These improvements concern the hard argmax decisions. Better ranking does not ensure that the leading category changes correctly for every lesion, so ranking and hard-decision results must be considered separately.

Mean per-class accuracy rises from 0.9267 to 0.9440, specificity from 0.9569 to 0.9638, PPV from 0.3233 to 0.6974 and NPV from 0.9533 to 0.9632. Specificity and NPV reflect the many negative

Table 4.1. Matched internal-validation comparison using the selected JI-ADF metric family. Values are proportions, not percentages.

Metric	Initial system	Enhanced ensemble
Macro ROC-AUC	0.8627	0.8767
Macro average precision	0.4560	0.5576
Mean per-class accuracy	0.9267	0.9440
Macro sensitivity	0.4282	0.5430
Macro specificity	0.9569	0.9638
Mean class Dice	0.3551	0.553
Macro PPV	0.3233	0.6974
Macro NPV	0.9533	0.9632

observations in the one-versus-rest tasks. They should be interpreted together with sensitivity and Dice, particularly for the smallest classes.

The enhanced representation combines information absent from a probability-only stack with image-view averaging. The observed gain concerns this complete combination and its fitting procedure. It does not establish that every component independently produces an improvement, nor that a new image backbone was learned during the frozen-feature study.

4.3.1 Counts Implied by the Reported Accuracy

For single-label argmax classification with $K = 11$ categories and $N = 786$ lesions, each misclassified lesion contributes one false positive and one false negative across the one-versus-rest tasks. Therefore, if E is the number of misclassified lesions,

$$\mathrm{Acc}_{\mathrm{OvR}} = 1 - \frac{2E}{KN}, \qquad E = \frac{KN}{2}\left(1 - \mathrm{Acc}_{\mathrm{OvR}}\right). \tag{4.5}$$

The calculated accuracies, rounded to four decimal places, uniquely imply the integer counts below under this evaluation rule. These counts are algebraically derived from the accuracies. The enhanced ensemble

Table 4.2. Lesion-level counts derived from the mean one-versus-rest accuracies.

Quantity	Initial system	Enhanced ensemble
Evaluated lesions	786	786
Correctly classified lesions	469	544
Misclassified lesions	317	242
Overall multiclass accuracy	59.67%	69.21%
Overall misclassification rate	40.33%	30.79%

has 75 fewer total errors, corresponding to a 9.54-percentage-point increase in overall multiclass accuracy. Aggregate counts do not identify which individual lesions changed prediction or how many initial errors were corrected, because some previously correct predictions may also have become incorrect.

Table 4.3. Literature values from JI-ADF Table 1 and our enhanced approach. Published benchmark rows and our internal-validation row use different evaluation cohorts.

Method	AUC	Precision*	Acc. (OvR)	Sens.	Spec.	Dice
JIF-MMFA (Tang et al., 2024)	0.750	0.328	0.896	0.406	0.939	0.306
VEMFL (Restrepo et al., 2024)	0.842	0.439	0.912	0.291	0.958	0.302
CAFFM (Tran-Van and Le, 2025)	0.787	0.306	0.898	0.233	0.954	0.232
ALBEF (Adebiyi et al., 2025)	0.760	0.294	0.889	0.208	0.945	0.222
CoscatNet-UFS (Zuo et al., 2025)	0.800	0.472	0.925	0.453	0.959	0.435
DualRefNet (Khurshid et al., 2025)	0.848	0.447	0.918	0.422	0.954	0.441
SkinM2Former (Zhang et al., 2025)	0.841	0.436	0.920	0.421	0.954	0.405
Fine-tuned PanDerm (Yan et al., 2025)	0.840	0.476	0.920	0.447	0.953	0.434
JI-ADF (Nguyen et al., 2026)	0.866	0.543	0.930	0.536	0.959	0.505
Enhanced EfficientNetV2 ensemble (ours)	**0.877**	**0.558**	**0.944**	**0.543**	**0.964**	**0.553**

4.4 Published JI-ADF Context

Table note. Literature results are reproduced from Table 1 of the JI-ADF paper and were not independently evaluated in this project.

Table 4.4. Published JI-ADF benchmark context and our validation.

Metric	JI-ADF paper	Enhanced ensemble
Macro ROC-AUC	0.866	0.8767
Macro average precision	0.543	0.5576
Mean per-class accuracy	0.930	0.9440
Macro sensitivity	0.536	0.5430
Macro specificity	0.959	0.9638
Mean class Dice	0.505	0.553
Macro PPV	0.596	0.6974
Macro NPV	0.960	0.9632

Our enhanced ensemble has numerically higher values for all eight listed metrics than the corresponding published JI-ADF values: macro ROC-AUC (0.8767 versus 0.866), macro average precision (0.5576 versus 0.543), mean per-class accuracy (0.9440 versus 0.930), macro sensitivity (0.5430 versus 0.536), macro specificity (0.9638 versus 0.959), mean class Dice / macro F1 (0.553 versus 0.505), macro PPV (0.6974 versus 0.596), and macro NPV (0.9632 versus 0.960). These numerical differences provide cross-study context; they do not establish superiority under a matched evaluation protocol.

Category recovery remains limited: macro sensitivity is 54.30% and mean class Dice / macro F1 is 0.553. A matching official test evaluation or a faithful reproduction under a shared protocol would be necessary to assess comparative generalization.

JI-ADF is a research reference and a separately implemented adaptation in this project. Our final method uses modular specialists and fixed geometric pooling, whereas JI-ADF uses joint–individual learning and adaptive decision weights. The comparison does not imply that our final system is a direct architectural extension of the paper.

4.5 Class-Level Evaluation

Small supports limit stability. The validation partition has one MAL_OTH lesion, seven VASC lesions and eight DF lesions. Their rates can change substantially after a single different prediction. Rank-based metrics, decision metrics and support counts should therefore be read together.

4.6 Inference Verification and the Demonstration

4.6.1 Checking a Wrong Nevus Prediction

A manually inspected holdout lesion, IL_0024228, has reference label NV but the initial score-fusion system predicts DF. Before attributing the mistake to training, we checked the deployed image preparation, metadata encoding, class order and specialist outputs against the preserved implementation [9].

The preprocessing fingerprint matched the original training configuration. For this case, the maximum difference between demonstration and preserved probability vectors was zero. The CPU-versus-MPS comparison differed by at most 7.58×10^{-9}. A selected 21-case parity sample also matched the archived predictions in all 21 accepted cases. These checks support numerical consistency for the tested paths; they are not an estimate of diagnostic accuracy.

The wrong NV-to-DF result was already present in the preserved model. It was therefore not created by a swapped class label or a newly changed demonstration transform in this investigation. We therefore studied class confusion and downstream fusion. Editing the interface label would have hidden the error.

4.6.2 Prediction and Explanation Contract

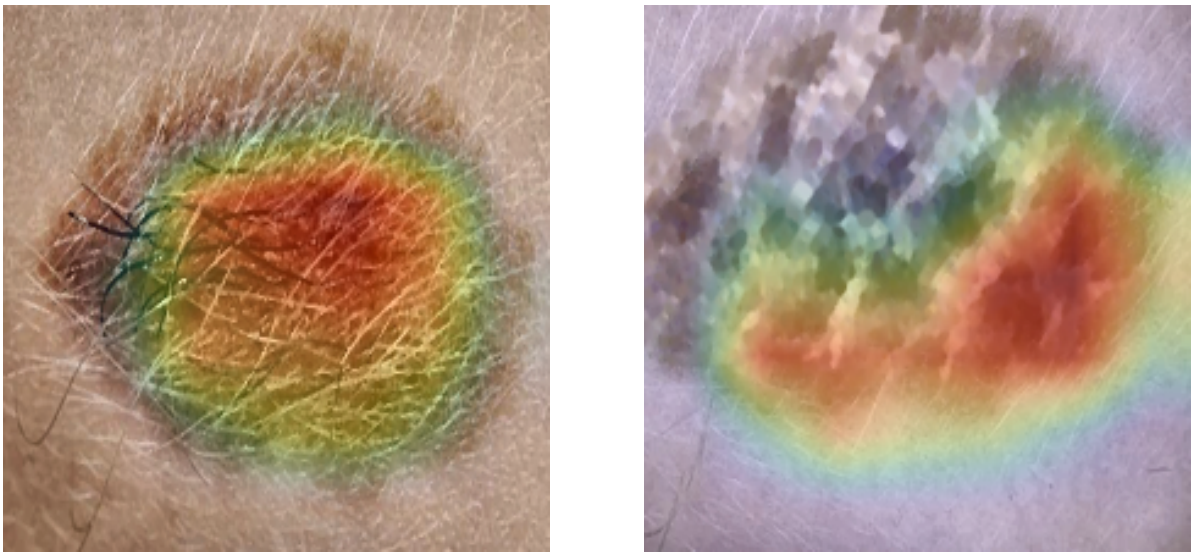
The prediction record retains the class code, complete score vector, image and metadata branch outputs, input-validation state and pipeline identity. The interpretation module checks vector dimensions, nonnegative finite values, normalization and consistency between the leading score and recorded category. The module does not generate a diagnosis summary for a malformed record.

Heatmaps are computed on the processed model input. The explanation function targets the final spatial feature layer, differentiates an image-branch logit and resizes the weighted activation map into the processed-image frame. Hooks are removed after computation, and the function does not update parameter gradients or model weights. If no spatially varying positive signal exists, the interface reports the explanation as unavailable.

4.6.3 Application Behaviour and Test Coverage

The test suite checks that rejection of either image occurs before disease preprocessing, that validator-only mode does not load disease models, and that paired fusion uses the training-time branch order. Other checks cover missing models, malformed scores, invalid artifact bundles and a heatmap failure after a valid prediction. The explanation tests use a controlled image pattern to verify spatial influence and check that model state stays unchanged. These are software tests; they do not substitute for testing classification on independent lesions.

The active application reads packaged downstream artifacts only after checking their hashes, class order, view list and encoder identities. Changing a checkpoint without its matching representation



(a) Clinical branch

(b) Dermoscopic branch

Figure 4.1. Branch-level Grad-CAM for the selected correct NV example IL_0008891. The case illustrates the interface and is not an additional evaluation cohort.

transforms raises an error. This protects against a common deployment mistake: fitting PCA on one encoder and later supplying features from another.

4.7 Interpretation of the Work

All selected internal-validation metrics improve over our initial score-fusion system. The paired-input architecture uses all three modalities while retaining the trained specialists. The corrected aggregates are numerically higher than the published JI-ADF values across all eight listed metrics, although sensitivity and class Dice remain limited.

Patient-level independence cannot be established from the available identifiers, and rare categories have small evaluation supports. Using the reference paper's metric terminology does not resolve these constraints. Broader-screening and supplementary binary performance results are outside this chapter's multiclass evaluation scope; their underlying prediction records and model artifacts remain unchanged.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusions

LesionLab was developed in two stages. We first implemented, adapted and trained a multimodal score-fusion system using clinical, dermoscopic and metadata specialists. We then enhanced our initial system by combining two compressed image representations, encoded metadata and specialist scores with predictions averaged across six views. The three trained specialists are retained while a new downstream decision layer is fitted.

On the same 786 validation lesions, the enhanced ensemble increases macro ROC-AUC from 0.8627 to 0.8767, average precision from 0.4560 to 0.5576 and mean class Dice from 0.3551 to 0.553. Mean per-class accuracy, sensitivity, specificity, PPV and NPV also improve over the initial score-fusion system. Mean per-class accuracy refers to one-versus-rest decisions and includes true negatives.

The published JI-ADF values provide a relevant metric framework, with different evaluation data and development conditions. Our corrected internal-validation values are numerically higher for all eight listed metrics. This is cross-study context rather than evidence of matched-test superiority.

The application adds relevance rejection, metadata handling, disagreement summaries and branch heatmaps. The parity investigation of a wrong nevus prediction showed why software correctness and diagnosis correctness require different checks.

5.2 Future Work

Future evaluation should lock an independent confirmation cohort before tuning, use patient grouping where identifiers are available and repeat specialist training across several seeds. These checks would assess whether the downstream improvement transfers beyond the current development configuration.

Rare-class work should review support counts and label quality before adding public data. New sources require modality, diagnosis and duplicate checks. Selective encoder fine-tuning should be compared with a frozen control under matched fitting access and selection criteria, with particular attention to melanoma and AKIEC sensitivity and class Dice.

Screening operating points should follow a declared sensitivity requirement, be selected on development data and remain fixed during external evaluation. Deployment work should measure latency and memory on the intended host, then test batching and fewer image views while tracking their effect on the selected classification metrics.

APPENDIX A

SUPPLEMENTARY BINARY STUDY DESIGN

The project contains a separate direct-binary study using MILK10k and ISIC 2019 for development and PAD-UFES-20 for external evaluation [10, 11]. Its endpoint, eligibility criteria and missing-modality handling differ from the eleven-class experiment. The report retains its design for implementation context, while restricting displayed performance results to the multiclass metric family used in the JI-ADF comparison.

A.1 Data and Endpoint

The binary branch defines eligibility from diagnostic labels before training. Clinical and dermoscopic inputs retain their modality identity, and an unavailable view remains absent. Shared metadata describe age, sex, coarse anatomical site and observation availability. Additional development dermoscopy comes from ISIC 2019. PAD-UFES-20 provides separately sourced clinical photographs.

Overlap checks quarantine groups with exact, identity or conservative perceptual similarity. Patient grouping is used when suitable identifiers are available. Dataset membership remains explicit so that external data do not silently enter development. Different diagnosis definitions require harmonization before samples can be pooled.

A.2 Candidate Design

The implementation compares frozen-feature classifiers, fixed fusion and reliability-aware combinations. Candidate selection and operating thresholds belong to the binary experiment rather than the eleven-class demo. These experimental branches use separate output directories and artifact identities.

A.3 Evaluation Separation

The external cohort has already been inspected, so later tuning would require an additional untouched confirmation cohort. The binary predictions, calibration settings and evaluation records remain available in the project, but their performance tables are outside this report's selected multiclass comparison. The active image specialists and saved baseline remain unchanged.

APPENDIX B

IMPLEMENTATION AND REPRODUCTION RECORD

B.1 Code Organisation

The project keeps model definitions, preprocessing, experiment drivers and application inference in separate modules. This appendix maps the report's descriptions to the relevant executable files, excluding temporary analyses and intermediate exports.

Table B.1. Implementation map for the main system.

Module or entry point	Responsibility
`prepare_data.py`, `make_splits.py`	Join labels and metadata, preserve image pairs and create the fixed lesion partitions.
`preprocessing.py`, `image_cache.py`	Prepare image pixels and reuse deterministic preprocessing.
`efficientnet_v2_image.py`	Construct the clinical and dermoscopic model instances and their eleven-class heads.
`metadata_mlp.py`	Define the twelve-input metadata classifier.
`train_kaggle.py`	Coordinate GPU training, resume states, logs and artifact export.
`feature_study.py`, `frozen_ensemble.py`	Build the frozen-feature candidates and implement the declared arithmetic or geometric combination.
`demo_ensemble.py`	Verify the selected artifacts and execute the frozen-feature/TTA ensemble.
`input_validation.py`	Handle image validity, relevance gating and rejection states.
`result_interpretation.py`	Summarize existing scores and flag ambiguity without changing the prediction.
`image_explanations.py`	Compute branch-specific Grad-CAM on the processed image.
`streamlit_app.py`	Provide upload, metadata, diagnosis, screening and evidence views.

Stage 1 of LesionLab is our implementation, adaptation and training of the initial EfficientNet, metadata and score-fusion system. Stage 2 enhances our initial system through frozen-feature fusion and view averaging. The project also adds relevance rejection, frozen-feature and view-averaging experiments, the interpretation layer, the interface and the screening and error analyses described in the report. The JI-ADF adaptation and the direct-binary study are separate experimental branches.

B.2 Notebooks and Runtime Separation

The main Kaggle notebook is `kaggle_training.ipynb`. The separate architectural adaptation uses `kaggle_ji_adf.ipynb`, and the supplementary binary experiment uses `kaggle_binary_v1.ipynb`.

Their output directories and checkpoint identities distinguish the experiments. Results from one experiment cannot be assigned to another model simply by starting its notebook.

The notebooks keep the attached dataset read-only and create writable project files under the Kaggle working directory. Archive extraction rejects unsafe paths. Split checks verify the expected training, validation, fusion and holdout labels before fitting. Resume configuration records the split fingerprint, modality and relevant optimization settings so that a resumed run cannot silently change the experiment definition.

B.3 Regenerating the Report's Main Tables

The report builder loads the preserved validation_scores.npz, including reference targets, lesion identifiers and both model score arrays. It computes one-versus-rest ranking and decision metrics for the eleven diagnostic categories. The paper-aligned tables use these calculated values directly; the historical evaluation records remain unchanged. The source ZIP includes the figure assets and bibliography needed to compile without the training datasets or model weights.

The numerical input has 786 unique lesion identifiers and score arrays of shape 786×11 for both methods. Replay checks enforce finite score values, compatible dimensions and the canonical category order. The artifact hashes in the evaluation record identify which prediction arrays and comparison files supplied the report. Regenerating a table does not rerun training or change a checkpoint.

B.4 Verification Boundaries

Application tests exercise rejection order, branch ordering, missing artifacts, malformed outputs and unavailable explanations. These tests establish properties of the implementation. The recorded 21-case parity check concerns agreement with the preserved inference path, while the 786-lesion evaluation concerns diagnosis outcomes. Neither test family substitutes for independent external validation.

The direct-binary study records its own external cohort, calibration thresholds and patient bootstrap. Appendix A retains its design without displaying binary performance results. This separation allows the work to be reproduced without interpreting an appendix result as an improvement in eleven-class classification.

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